

AI-Enabled Mobile X-ray System for Modern Healthcare

DRGEM RAYMO

1. Introduction to AI Mobile X-ray System

An AI Mobile X-ray System is designed to provide fast, accurate, and portable X-ray imaging at the patient's location, while using AI to improve image quality and support the medical imaging workflow.



Fig. 1 – AI-enabled mobile digital radiography unit

1.1 Introduction

An AI Mobile X-ray System is a portable digital radiography device that allows healthcare professionals to capture X-ray images directly at the patient's bedside. Unlike traditional X-ray machines that require patients to be transported to the radiology department, mobile X-ray systems can be moved easily to emergency rooms, intensive care units (ICUs), operating rooms, isolation wards, and ambulances.



Fig. 2 – Bedside imaging in a hospital ward

Artificial Intelligence (AI) has transformed modern mobile X-ray systems by improving image quality, reducing examination time, assisting radiologists in detecting abnormalities, and minimizing radiation exposure. These intelligent systems are particularly valuable for critically ill patients who cannot be moved safely.

The DRGEM RAYMO is one of the latest AI-enabled mobile digital radiography systems. It combines compact hardware, wireless operation, intelligent positioning assistance, RFID-based user personalization, and AI-supported workflow features to improve efficiency and patient safety. It was recently showcased at India Health 2026 and has received regulatory approvals for global markets.

Source: DRGEM

1.2 Need for AI Mobile X-ray Systems

Modern hospitals require fast and accurate diagnostic imaging because:

- ICU patients cannot be transported easily.
- Emergency diagnosis must be performed within minutes.
- Infection control requires imaging at the patient's location.
- Rural hospitals need portable imaging equipment.
- Disaster response requires mobile diagnostic systems.

AI-based mobile X-ray systems solve these challenges by enabling bedside imaging with intelligent software assistance.

1.3 Objectives of AI Mobile X-ray System

The primary objective of an AI Mobile X-ray System is to provide fast, safe, and accurate diagnostic imaging directly at the patient's location without the need to transport them to the radiology department. By combining digital radiography with Artificial Intelligence (AI), the system enhances image quality, improves workflow, reduces human error, and supports healthcare professionals in making faster clinical decisions. It is especially useful in emergency departments, intensive care units (ICUs), operating rooms, isolation wards, and remote healthcare facilities where immediate imaging is essential. The integration of AI also enables automated image optimization, assists in detecting abnormalities, and reduces the time required for diagnosis. Overall, the system improves patient care, enhances hospital efficiency, and contributes to better healthcare outcomes.

Objectives:

- To provide high-quality digital X-ray imaging at the patient's bedside.
- To eliminate the need to move critically ill or injured patients.
- To improve the speed and accuracy of medical diagnosis.
- To reduce radiation exposure using AI-based dose optimization.
- To enhance image quality through AI-powered image processing.
- To support radiologists by identifying possible abnormalities automatically.
- To reduce examination time and increase patient throughput.
- To improve workflow efficiency in hospitals and emergency departments.
- To enable wireless image transfer through PACS and DICOM networks.
- To minimize repeat X-ray examinations by producing clear images on the first attempt.
- To reduce human errors through intelligent automation.
- To provide easy operation with a touchscreen interface and automated controls.
- To improve infection control by performing imaging at the patient's location.
- To support telemedicine by allowing remote image review and consultation.
- To deliver reliable diagnostic services in rural and mobile healthcare environments.

1.4 Key Characteristics of AI Mobile X-ray System

An AI Mobile X-ray System is designed with advanced technologies that make diagnostic imaging faster, more accurate, and highly efficient. Unlike conventional X-ray machines, it is lightweight, portable, battery-powered, and equipped with intelligent software that automatically enhances image quality and simplifies the imaging process. The system allows healthcare professionals to perform examinations directly at the patient's bedside, making it ideal for emergency care, ICUs, operation theatres, and isolation wards. Its wireless connectivity, AI-based image enhancement, cloud integration, and digital workflow significantly improve hospital productivity while ensuring patient safety and comfort.

Key Characteristics:

- Portable and compact design for easy transportation.
- Battery-powered operation for continuous use without external power.
- High-resolution digital flat-panel detector.
- AI-assisted image enhancement for better diagnostic quality.
- Automatic exposure detection and image optimization.
- Touchscreen control panel with a user-friendly interface.
- Wireless detector for flexible positioning.
- DICOM compatibility for seamless communication with hospital systems.
- PACS integration for secure image storage and retrieval.
- Fast image acquisition and instant display.
- AI-assisted positioning guidance for accurate imaging.
- Low radiation dose while maintaining excellent image quality.
- RFID or user authentication for secure access.
- Built-in driving camera to improve navigation in crowded hospital areas.
- Remote operation using a wireless tablet or mobile device.
- Cloud connectivity for remote diagnosis and tele-radiology.
- Automatic quality control and system diagnostics.
- Easy maneuverability with motor-assisted movement.
- Suitable for adult, pediatric, neonatal, and emergency patients.
- Reliable operation in ICUs, emergency rooms, ambulances, and field hospitals.

1.5 Importance of AI Mobile X-ray System in Modern Healthcare

AI Mobile X-ray Systems have become an essential part of modern healthcare because they enable healthcare professionals to perform diagnostic imaging quickly and efficiently without transporting patients. This capability is especially valuable for critically ill patients, trauma victims, elderly individuals, newborn babies, and patients with infectious diseases. The integration of Artificial Intelligence improves image quality, reduces radiation exposure, assists in detecting abnormalities, and speeds up diagnosis. Hospitals benefit from faster workflows, fewer repeat examinations, and better utilization of radiology resources. These systems also support telemedicine by allowing images to be shared instantly with specialists located anywhere in the world. As healthcare increasingly adopts digital technologies, AI Mobile X-ray Systems play a vital role in improving patient safety, reducing operational costs, and delivering high-quality medical care.

Importance:

- Enables immediate bedside diagnosis for critically ill patients.
- Reduces the need to transport patients to the radiology department.
- Saves valuable time during emergencies and trauma care.
- Improves diagnostic accuracy through AI-assisted image analysis.
- Produces high-quality digital images within seconds.
- Reduces radiation exposure for patients and healthcare workers.
- Minimizes repeat imaging through intelligent image optimization.

- Improves patient comfort and safety.
- Enhances workflow efficiency in hospitals.
- Supports intensive care units (ICUs) and emergency departments.
- Helps manage infectious disease patients by reducing patient movement.
- Supports tele-radiology and remote specialist consultation.
- Increases productivity of radiologists and technicians.
- Reduces healthcare costs through faster diagnosis and fewer repeat scans.
- Provides reliable imaging services in rural, mobile, and disaster-relief settings.
- Improves overall quality of healthcare delivery.
- Contributes to early disease detection and timely treatment.
- Supports digital transformation and smart hospital initiatives.
- Enhances clinical decision-making using AI-powered insights.
- Represents the future of portable diagnostic imaging technology.

2. History and Evolution of Mobile X-ray



Fig. 3 – Modern AI-enabled mobile X-ray unit

2.1 Early Development

The first X-ray machine was invented by Wilhelm Conrad Röntgen in 1895. Early systems were very large, required film processing, and were permanently installed in radiology departments.

During World War I, portable X-ray machines were developed to diagnose injured soldiers in field hospitals. These early systems were heavy and had limited imaging capability.

2.2 Transition to Digital Radiography

Medical imaging evolved through several stages:

Year	Technology
1895	Film X-ray
1970	Computed Radiography (CR)
2000	Digital Radiography (DR)
2015	Wireless Mobile DR
2020	AI-assisted Digital Radiography
2026	Intelligent AI Mobile X-ray Systems (e.g., DRGEM RAYMO)

2.3 Evolution of Mobile X-ray Technology

First Generation

- Film-based imaging
- Heavy equipment
- Manual operation

Second Generation

- Digital detectors
- Better image quality
- Faster processing

Third Generation

- Wireless detectors
- Battery-powered systems
- PACS integration

Fourth Generation (Current)

Modern systems include:

- Artificial Intelligence
- RFID login
- Live driving camera
- Wireless tablet control
- Automatic image optimization
- Cloud connectivity
- Real-time image processing
- Smart workflow automation

These features reduce operator workload while improving diagnostic efficiency.

Source: DRGEM

2.4 Why Hospitals are Replacing Older Systems

Hospitals are adopting AI-enabled mobile X-ray systems because they provide:

- Faster bedside imaging
- Better image quality
- Lower radiation dose
- Reduced repeat examinations
- Improved patient safety
- Better ICU workflow
- AI-assisted clinical decision support

Latest News (2026)

- India Health 2026 (New Delhi): DRGEM showcased its latest mobile X-ray portfolio, including RAYMO, PROMO, and TOPAZ. RAYMO attracted attention for its AI-enabled workflow, compact design, and bedside imaging capabilities. (Express Healthcare)
- European Congress of Radiology (ECR 2026): DRGEM introduced the next-generation RAYMO, featuring a design about 20% smaller than previous models, an integrated driving camera, RFID-based user personalization, and wireless tablet control for improved mobility and workflow. (DRGEM)
- Global Expansion: DRGEM's RAYMO recently received U.S. FDA clearance and European CE certification, supporting wider international adoption. (DRGEM)